Selected ambient cuts and three prefix orbits

Human Compression project: L044 continuation

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle$ act transitively on Ω, with involutory marks and string commutations. Eligibility always means crossing two orbits of the actual full ambient omission G_j. The first lemma extends the acquired ambient interval proof to selected colours inside a larger interval. The subgroup whose orbits are counted retains *all* colours of that larger interval.

Lemma 1 (Selected ambient confinement)**.** *Let $I = \{u, \dots, v\}$, $T = G_I$ and $F \subseteq I$. Suppose V is a proper T-invariant subset of Ω. For every $j \in F$, suppose at least two ambient-eligible j-edges exist and every such edge has both endpoints in V. Then, including singleton T-orbits in the count,*

$$2|F| \leq |V| - o_T(V). \tag{1}$$

The set F need not be an interval, and marks in $I \setminus F$ need not be trivial on V.

Proof. If F is empty the assertion is immediate. Otherwise choose two eligible edges per colour in F on the full vertex set V. The complete acquired selected-colour square-normal-form proof uses the actual ambient omissions just as it uses full local omissions: every alternative path avoiding a colour lies in its ambient omission, and conjugation by any other mark preserves its orbit partition. Thus long-cycle reduction, incident-square merging, ordinary slides and global middle moves remain available. All newly chosen edges remain in V by the all-edge confinement hypothesis. A middle move for a square of colours $i-1, i+1$ uses $u < i < v$ and hence $\rho_i \in T$, which preserves V. The selected-colour proof explicitly permits this middle colour to be unselected: retain the square and transport the other selected edges by ρ_i; there are simply no selected i-edges to account for when $i \notin F$.

Choose a square-only selection with minimum square count q. The paid minimum-distance argument therefore gives at most one square per component in every such minimum selection. Each component is a tree or is unicyclic. All selected edges lie inside actual T-orbits. We prove that each nontrivial orbit $O \subseteq V$ has a tree component, by checking the full acquired ambient-orbit exit argument with the changed colour set.

Suppose a minimum selection has no tree in O; its selected edge count is $E_O = |O|$. A square colour is localized to its square's component: an eligible edge inside another component would have an alternative selected path avoiding that colour, while an edge between components would replace a square edge and decrease q. The complete ordinary-slide proof consequently still preserves component vertex sets. It uses only selected bridge and square colours, all in F, and is independent of missing other colours.

Every full neighbour of a square vertex stays in its component. Neighbours of a square colour already lie in the square. If the neighbour's colour commutes with a square colour, transport that eligible square edge and use the localization just proved. The only remaining case is square colours $i-1, i+1$ and neighbour colour i. Here $\rho_i \in T$ preserves each actual orbit, including O, even if $i \notin F$. The global move preserves $E_O = |O|$. Viewed on the transported selection, it removes the transported square and inserts the original square. Removal splits the transported component into four trees. An inserted vertex cannot lie in another component of O, since that would join two squares, nor outside O, which ρ_i preserves. All four trees must be rejoined, since otherwise a tree would remain in O despite $E_O = |O|$. Thus the component partition inside O is transported by ρ_i, and the required neighbour lies in the original component. This checks the

acquired partition argument with an unselected middle mark; tree components in other T-orbits cannot interfere.

Because O is proper and G is transitive, a full edge leaves O. Over all minimum selections with $E_O = |O|$, choose a shortest connector consisting of a selected path from a square to the inside endpoint of such an edge, followed by that exit edge. Its colour x is outside I, since every mark in I preserves O. The selected path is nonempty by the neighbour assertion. If its first colour commutes with a square colour, the acquired ordinary slide shortens the path and preserves the exit. Its repeated-colour exception is excluded by square-colour localization: as in the full ambient-orbit proof, that exception would place the completing eligible square edge outside the original component. Every colour used by this slide is selected, so missing colours introduce no new case.

The remaining first colour is i, with square colours $i-1, i+1$. Here $u < i < v$ and $x \notin I$ give $|i - x| \geq 2$. The global middle move therefore transports the exit to an actual x-edge. It remains an exit from O, because $\rho_i \in T$ preserves O and its complement. The transported selected path starts on the retained square after its first edge has disappeared; it gives a shorter connector, still with $E_O = |O|$, a contradiction.

Thus every nontrivial T-orbit in V has a tree. Each singleton is an isolated tree as well. Trees contribute one to vertices minus edges and unicyclic components contribute zero. Summing gives (1). The proof used confinement of every eligible selected colour edge, not merely the original choices. □

Theorem 2 (Three moving orbits)**.** *Retain the complete critical singleton and small-support budget hypotheses. Write $P = G_{<g}$, $Y = \mathrm{supp}(P)$, $H = G_{\{2,\dots,g-1\}}$, $C = a^P$ and $b = a^{\rho_0}$, where $4 \leq g < r$, $n = 2r \geq 14$, P is intransitive on Y, $|Y| \leq 2g+1$, and every colour $2, \dots, g-1$ has two ambient-eligible edges. Then H cannot have exactly two fixed points and three nontrivial orbits in Y.*

Proof. Suppose it does. The acquired budget gives $|Y| = 2g+1$, fixed points exactly a, b, and three nontrivial orbits whose total size is $2g-1$. The centralizing involution ρ_0 acts on the fixed set as $(a\ b)$ and permutes those three orbits equivariantly.

Suppose first that it exchanges two orbits U_1, U_2 of equal size. The third orbit W has odd size and is preserved, so ρ_0 fixes it pointwise. Strict growth makes $|C| \geq g+1 > 2$, so C meets a moving H-orbit. If it meets either U_i, it contains both. Intransitivity of P then forces $C = \{a, b\} \cup U_1 \cup U_2$, the support of ρ_0. Since ρ_1 preserves C and every later mark centralizes ρ_0, this would make G intransitive.

Otherwise $C = \{a, b\} \cup W$ and its complement in Y is $E = U_1 \cup U_2$. These are the two P-orbits. The involution ρ_0 is free on E and fixes W. Strict growth gives $x = b^{\rho_1} \in W$, so the first-prefix anchor orbit is $\{a, b, x\}$. The actual natural singleton stabilizer and full IP give $|P_1| = 6$, for $P_1 = \langle \rho_0, \rho_1 \rangle$. The relation $(\rho_0\rho_1)^3 = 1$ makes P_1 fix $W \setminus \{x\}$ pointwise. Its orbits in E have sizes two or six, and these are all its ambient orbits of those sizes. All marks of index at least three centralize P_1 and preserve their union E. The first three marks preserve E as well, since $\rho_2 \in H$. This again contradicts transitivity.

Thus ρ_0 preserves each of the three H-orbits. On each it is either free or the identity. At least one is fixed pointwise, since their total size is odd. At least one is free, since otherwise $\rho_0 = (a\ b)$ would be odd. Write the free orbits as $O_1, \dots, O_k$, where $k = 1$ or 2 and $|O_i| = 2q_i$. Let V be the union of the other $t = 3 - k$ orbits, of total size M. Thus

$$2\sum_i q_i + M = 2g - 1. \tag{2}$$

Assign to one free orbit every colour in $I = \{2, \dots, g-1\}$ which has an ambient-eligible edge in any free orbit. Let F_i be the disjoint assigned sets, and let F consist of the remaining colours. The acquired internal-pair forest argument gives

$$|F_i| \leq q_i - 1. \tag{3}$$

For clarity, its first part needs no terminal-interval hypothesis: project one chosen eligible edge of each assigned colour to the ρ_0-pairs in O_i. Its endpoints are distinct pairs, since $\rho_0 \in G_j$. An alternate projected path avoiding colour j lifts using the other chosen colours and ρ_0, all in G_j, and contradicts eligibility. Hence these distinct-coloured projected edges form a forest on q_i vertices, proving (3). All marks in the actual interval H preserve O_i and its pairs; no projected IP or stronger equality classification is invoked.

For every $j \in F$, all its ambient-eligible edges lie in V: none lies in a free orbit by the definition of F, and the other points are fixed by H. The set V is proper and H-invariant, with exactly t actual H-orbits. Lemma 1, with the full interval I and its selected subset F, gives

$$2|F| \leq M - t. \tag{4}$$

This also holds when F is empty. Summing the disjoint colour assignments and using (2)–(4), we obtain

$$\begin{aligned} g - 2 &= \sum_i |F_i| + |F| \\ &\leq \sum_i (q_i - 1) + \frac{M - t}{2} = \frac{2g - 1}{2} - k - \frac{3 - k}{2} = g - 2 - \frac{k}{2} < g - 2, \end{aligned}$$

the final contradiction. □

Attribution and exact boundary. The selected-colour normal form, full ambient-orbit localization and interval exit proof are acquired arguments; their historical global middle move retains the root collaborator's disclosed contribution and distinct review. Lemma 1 combines their exact scopes by retaining the full interval subgroup while selecting fewer colours. The confinement and pair-forest budget in Theorem 2 are this continuation. The source setting is Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, Propositions 5.28–5.32. This excludes the stated third-orbit configuration. It does not itself assemble the entire singleton route or settle the separate imprimitive, all-paired, finite-base and lower-existence obligations of the principal L044 contract. No finite census, new formal declaration or whole-original credit is added.